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Work Problems

In some ways, work problems are like motion problems. In a motion problem, you know that the *rate* of travel multiplied by the *time* traveled is equal to the *distance*: $\text{rate} \times \text{time} = \text{distance}$

In a work situation, the *rate* at which someone works multiplied by the *time* worked is equal to how much *work* can be done: $\text{rate} \times \text{time} = \text{work}$

Work problems often use fractions because the time worked is enough to complete only a fraction of the job. For example, a mover estimates that he can completely load a truck in 4 hours. After 1 hour, he should have $\frac{1}{4}$ of the job done.

Example 1

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Samuel is a house painter. He can finish a job in 6 hours. Working alone, his assistant Scott could finish the job in 8 hours. If they work together, how long will the job take? The problem is about 2 people who work at different rates. When they work together, they will work the same amount of time, even though one is working faster.

Step 1

Use a chart.

If it takes Samuel 6 hours, he can finish $\frac{1}{6}$ of the job in 1 hour.

If Scott can do the job in 8 hours, he can finish $\frac{1}{8}$ of the job in 1 hour.

	Rate: Fraction of Job per Hour	Time Worked	Work Done per Time Worked
Samuel	$\frac{1}{6}$	x	$\frac{1}{6}x$
Scott	$\frac{1}{8}$	x	$\frac{1}{8}x$

Let x be the time worked in hours.

$\rightarrow \left(\frac{1}{6}\right)x$

$\rightarrow \left(\frac{1}{8}\right)x$

Step 2

Write an equation. The equation is set equal to 1 whole job.



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$$\frac{1}{6}x + \frac{1}{8}x = 1 \quad \longrightarrow \times 24 \text{ on both sides:}$$

Find a common denominator and add.

$$\frac{4}{24}x + \frac{3}{24}x = 1$$

$$\frac{7}{24}x = 1$$

Multiply by the reciprocal.

$$\frac{24}{7} \times \frac{7}{24}x = 1 \times \frac{24}{7}$$

Change the answer to a mixed number.

$$x = \frac{24}{7}, \text{ which equals } 3\frac{3}{7} \text{ hours}$$

The job will take $3\frac{3}{7}$ hours.

$$24\left(\frac{1}{6}\right)x + 24\left(\frac{1}{8}\right)x = 24(1)$$

$$4x + 3x = 24$$

$$7x = 24$$

$$x = \frac{24}{7}$$

$$x = 3\frac{3}{7} \text{ h.}$$

It makes sense that by helping each other, Samuel and Scott should be able to do the work faster. Always use common sense to check your work.

Example 2

[Hide](#)

Working alone, either Karla or her son could clean out the garage in 10 hours. They decide to work together, but her son has to leave after 3 hours, and Karla finishes by herself. How many more hours does Karla work after her son leaves?

Step 1

Let x represent the number of hours worked.

$$\text{Karla's work per hour: } \frac{1}{10}x$$

$$\text{Her son's work per hour: } \frac{1}{10}x$$

$$\text{After 3 hours, they have finished } \frac{3}{10} + \frac{3}{10} = \frac{6}{10} \text{ of the job.}$$

$$\text{Only } \frac{4}{10} \text{ remains.}$$

Step 2

Write an equation to figure out how long it will take Karla to

$$\text{do the final } \frac{4}{10} \text{ of the job.}$$

$$\frac{1}{10}x = \frac{4}{10}$$

$$x = 4$$

Karla will need to work 4 more hours after her son leaves.

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